Supporting Information

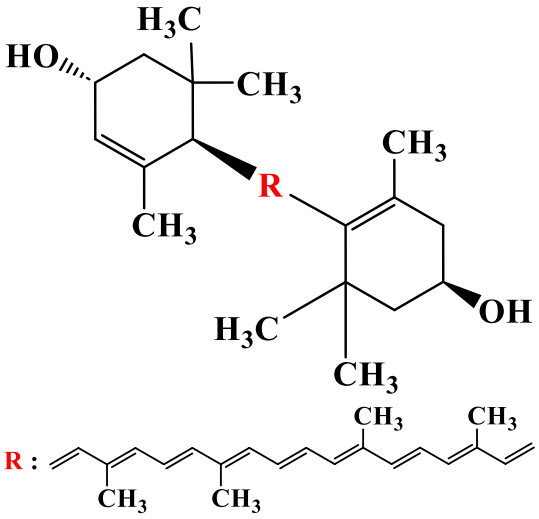
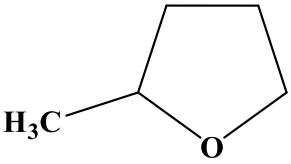
Optimizing Solvent Systems for Effective Lutein Solvation: A Molecular Dynamics Approach Using 2-Methyl Tetrahydrofuran, Deep Eutectic Solvent, Microemulsion, and Hexane

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Table S1. Structure and composition of lutein and solvent systems

Component category	Component name	Molecular structure	Composition (wt%)
solute	Lutein		-
Green Solvent	2- methyl tetrahydrofuran		-

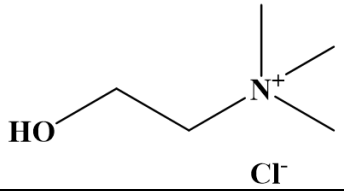
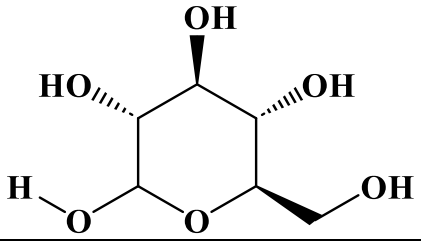
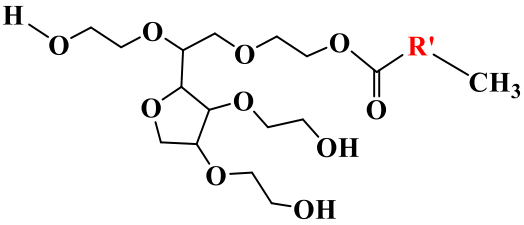
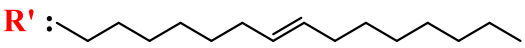
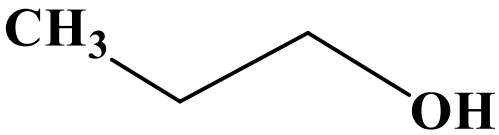
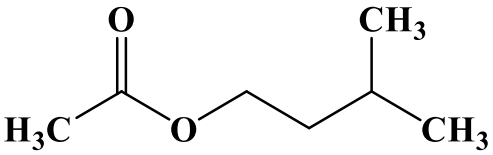
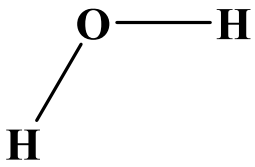
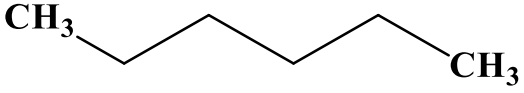
Deep Eutectic Solvent	Choline chloride		30.08
	Glucose		69.92
Microemulsion	Tween80	 	36
	n-propanol		36
	Isoamyl acetate		18
	Water		10
Conventional Solvent	n- Hexane		-

Table S2. Density of solvents

System Name	Simulated Density (Kg/m^3)	Experimental Density (Kg/m^3)	References
DES	1354.87	1300	[1]
2-Methyltetrahydrofuran	878.28	842.8	[2]
Hexane	651.35	654.8	[3]

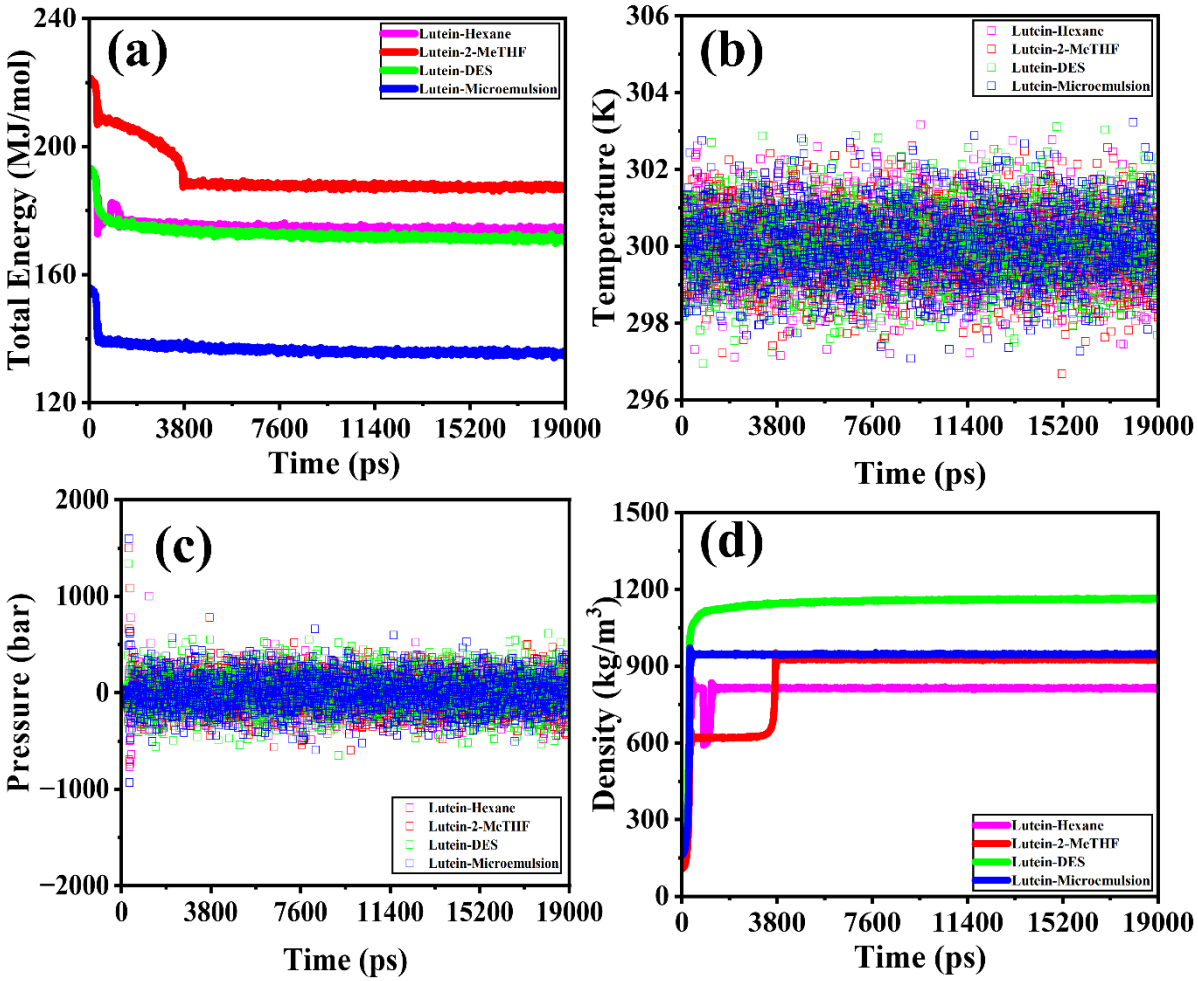


Figure S1. Equilibration metrices (a) Total Energy (b) Temperature (c) Pressure (d) Density

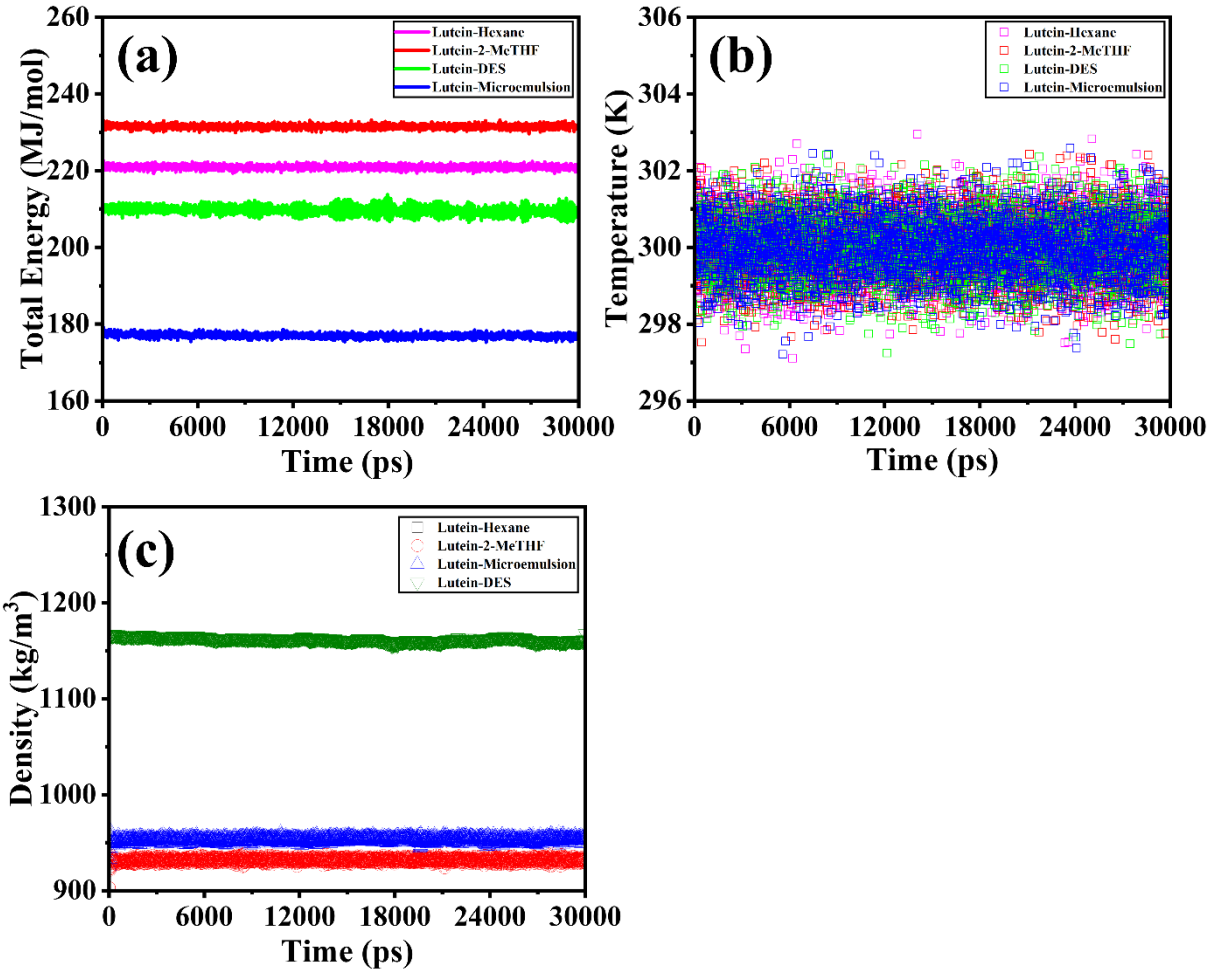


Figure S2. Production run metrices (a) Total Energy (b) Temperature (c) Density

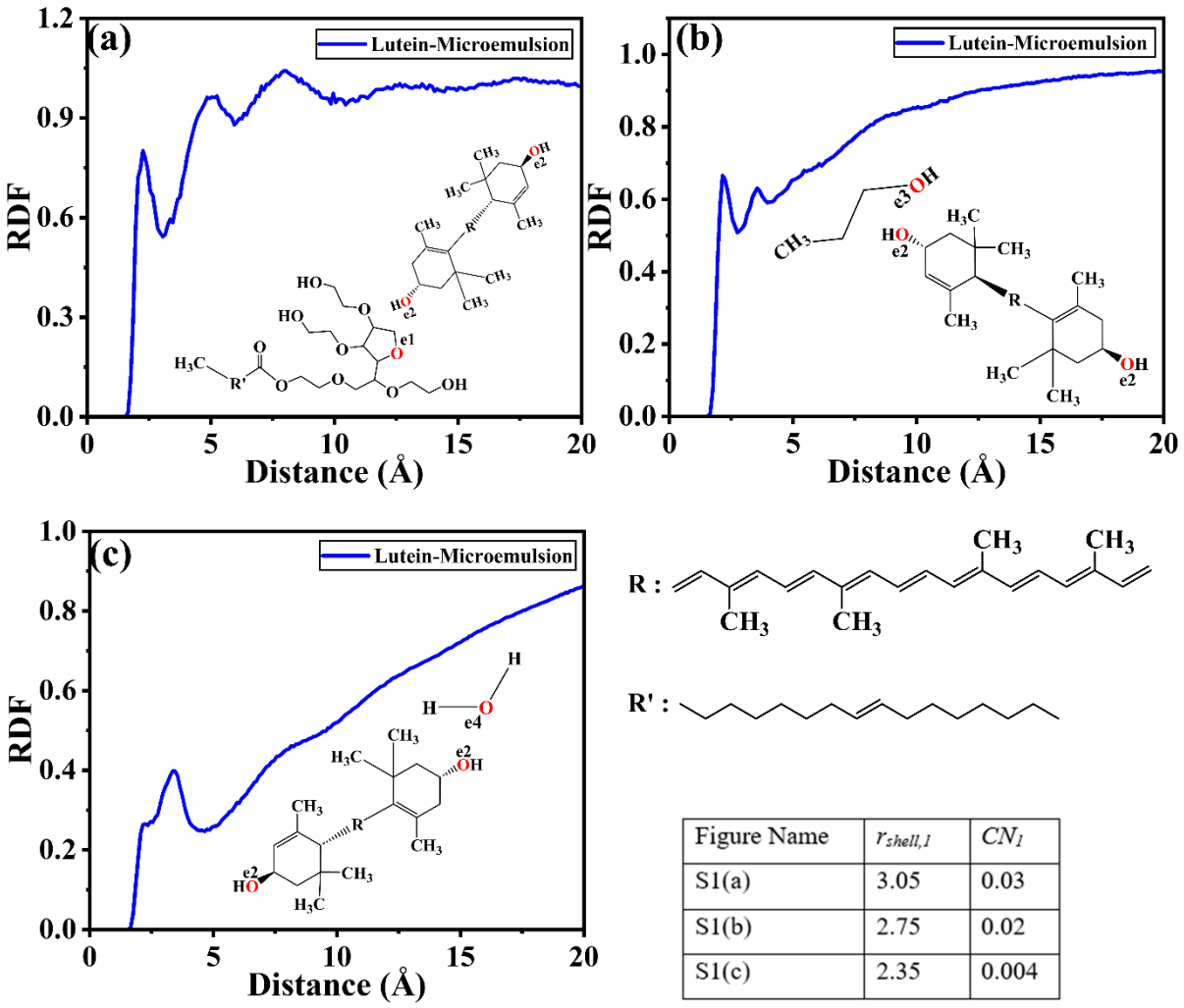


Figure Name	$r_{shell,1}$	CN_1
S1(a)	3.05	0.03
S1(b)	2.75	0.02
S1(c)	2.35	0.004

Figure S3. RDF graph of interaction between lutein and components of Microemulsion (a) O (heterocyclic moiety of tween 80)- H (hydroxyl moiety of lutein), (b) O (hydroxyl moiety of n-propanol)- H (hydroxyl moiety of lutein), (c) O (water)- H (hydroxyl moiety of lutein).

References

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[2] F. Comelli, R. Francesconi, A. Bigi, K. Rubini, Molar Heat Capacities, Densities, Viscosities, and Refractive Indices of Dimethyl Sulfoxide + Tetrahydropyran and + 2-Methyltetrahydrofuran at (293.15, 303.15, and 313.15) K, J. Chem. Eng. Data 52(2) (2007) 639-644. https:// doi.org/10.1021/je600557n.

[3] K. Lal, N. Tripathi, G.P. Dubey, Densities, Viscosities, and Refractive Indices of Binary Liquid Mixtures of Hexane, Decane, Hexadecane, and Squalane with Benzene at 298.15 K, J. Chem. Eng. Data 45(5) (2000) 961-964. https:// doi.org/10.1021/je000103x.